ARGUS: Learning a Bird from Five Calls

Two-Stage Few-Shot Detection of a Threatened Western Ghats Endemic, and a Measurement Gap in How the Field Scores It

IRIS National Fair 2026: Project Synopsis

Students: Subham Sai Samal, Vikramzharis | Guide: Mrs. Valarmathy, Teacher
Category: Earth & Environmental Sciences | Code and data: github.com/SubhamSaiSamal/ArgusSite

1. Abstract

The Western Ghats is one of Earth's eight hottest biodiversity hotspots, yet one of the least acoustically monitored. Deep-learning detectors usually need hundreds to thousands of labelled examples per species, which threatened species lack. *Sholicola albiventris*, an endemic bird listed as IUCN Near Threatened, has only 22 labelled recordings in BirdCLEF 2024. The species conservationists most need to track are data-poor for the same reason they are threatened.

Our goal was a system that learns a new species from five example calls and tells it apart from similar-sounding species better than chance. We built a two-stage propose-then-verify pipeline. A 445,000-parameter convolutional network, trained from scratch with a prototypical objective, proposes candidate calls. A second stage embeds each candidate with a pretrained bird-audio model and classifies it with logistic regression trained on the same five examples.

Across ten Western Ghats species, each tested against its four nearest acoustic confusers, the first stage alone reached a mean species-discrimination AUC of 0.585, barely above chance. Verification improved all ten species, to a mean AUC of 0.751. This gain is 5.2 times the measured noise floor (0.032), held across four independent runs, with four times fewer false alarms.

We also found that the standard detection test, which mixes in no confusers, barely changed (average precision 0.611 to 0.604) while discrimination against the nearest confusers rose from 0.585 to 0.751, because that test never checks whether the right species was identified. This two-stage approach offers a first screening step for data-poor birds.

2. Introduction & Objective

Passive acoustic recorders can survey a forest continuously and cheaply, but the detectors that process their audio usually need hundreds to thousands of labelled examples per species. This creates a bottleneck: the species that most need monitoring are the ones with the fewest recordings. *Sholicola albiventris*, endemic to the Western Ghats sky islands and IUCN Near Threatened, has only 22 labelled recordings in BirdCLEF 2024, compared with about 500 for common species.

Existing few-shot bioacoustic work mainly asks whether a system can detect an event. For a conservation record, a detection is only useful if it is the right species, especially in forests full of similar-sounding birds.

Objective: to build and test a system that identifies a species well above chance from five example calls, measured against its most similar-sounding confusers and an empirically measured noise floor.

3. Innovation

This project makes three contributions. First, a propose-then-verify design that separates finding calls from identifying the species: a small network trained from scratch finds candidate calls, and a separate embedding-based stage decides which species made each one. Second, a harder test: every species is scored against its four nearest acoustic confusers, not against easy random species. Third, we show that the standard detection score (AP) cannot see species mix-ups. Ranked by AP, our verified pipeline finishes last of six fusion rules; ranked by target-versus-confuser ROC-AUC, it finishes first.

4. Methodology

Data: BirdCLEF 2024. Ten Western Ghats target species were each tested against their four nearest acoustic confusers. Targets, confusers and evaluation metrics were fixed in advance to prevent cherry-picking.

Stage 1 (proposal): a 445,000-parameter convolutional neural network was trained episodically on 40 other species (none of the 10 targets) with a prototypical-network objective on PCEN spectrograms. At test time it is fine-tuned for 250 steps on the five support calls and local negative samples.

Stage 2 (verification): surviving candidates are embedded with a pretrained bird-audio model (Perch v2) and scored by a logistic-regression verifier fitted on the same five support calls.

Evaluation: target calls were planted into real Western Ghats soundscapes at controlled signal-to-noise ratios, with calibration and evaluation recordings kept separate. We measured the run-to-run noise floor at 0.032 by repeating identical runs, and only counted effects larger than this.

Controls: false alarms were counted on target-absent soundscapes. We also tested three alternative explanations: the pretrained model working alone, the model learning the recordist's equipment (using a recordist-disjoint split), and stage 1's choice of negative examples.

5. Results & Conclusions

Proposal-only matching separated target calls from their nearest confusers at a mean AUC of 0.585 across ten species, at or below chance for two. Adding verification improved every species, to a mean AUC of 0.751. The gain is 5.2 times the measured noise floor, was reproduced in four independent runs, and false alarms on target-absent audio fell from 54 to 13 per hour.

Four published techniques, two from challenge-winning systems, failed to beat our baseline. Compressing the verifier into the small network lost 0.136 AUC and fell to chance for the study species, a reported negative result. On 26.6 hours of unlabelled forest audio, the study species was flagged 1.74 times as often as a control species; expert review is pending.

Conclusion: five calls carry enough information to tell a threatened species apart from its nearest confusers well above chance, offering a practical screening filter for data-poor conservation.

6. Acknowledgement & References

Guided by our teacher, Mrs. Valarmathy. We thank Chiti Arvind (IISER Tirupati) for feedback on vocal variation in *Sholicola albiventris*. We used BirdCLEF 2024 and Google's Perch v2 on free public GPUs. AI tools (Claude, Antigravity) wrote much of the code and helped draft this text; all experiments and results are our own.

References: Nolasco et al., *Learning to detect an animal sound from five examples*, Ecol. Inform., 2023. Kahl et al., *Overview of BirdCLEF 2024*, 2024. van Merriënboer et al., *Perch 2.0*, 2025. Snell et al., *Prototypical Networks*, NeurIPS, 2017.